

Parley

Agents that **negotiate the price**, then pay it.

A buyer and a seller strike deals inside owner-set guardrails and settle in USDC on Arc.

Track: Agentic Economy

github.com/edwardparker8760/parley

125 tests green

Agent payments exist. Agent *pricing* does not.

Today: fixed-price payments

Existing agentic-payment demos, including Circle's own nanopayments starter, show an agent paying a **fixed, posted price**. The economically interesting step, deciding *what the price should be*, never happens.

Parley: price discovery

Two agents with **opposing interests** negotiate price, quantity and terms on behalf of their owners. Settlement is the **last step, not the product**.

If agents are going to transact for us, they must be able to disagree, concede, and walk away, without ever breaking their owner's limits.

Owner limits are arithmetic, not instructions

Deterministic layer: correctness

- Utility functions and a back-loaded concession schedule
- ZOPA inferred from revealed offers only
- **Hard guardrail clamp** computes the feasible band; nothing outside it can be sent
- An **independent egress guard** on the bus re-derives the band and rejects breaches

LLM layer: judgement and legibility

- Chooses *where inside a 2% window* to land
- Writes the rationale attached to every offer
- Schema-validated; an out-of-band pick is discarded, its words kept
- Timeout, transport error and bad schema all fall back deterministically

*The band is computed **before** the model is consulted and re-checked **twice** after. A fully compromised model still cannot produce an out-of-band offer.*

The safety claim, mechanically checked

0

out-of-band offers on the wire from a model answering **99999999** to every prompt, across all three scenarios. Every refusal is logged with the number arithmetic threw away.

0

settlement calls on a walk-away. Unreachable by construction, asserted by a counting-spy test over scenario C, against exactly 1 for scenario A.

125

tests: band properties over an adversarial corpus, prompt injection through the counterparty's own text, prompt-leak in both directions, byte-identical replay from SQLite.

*Swapping the LLM provider mid-build changed **zero** tests in the injection suite. The safety property belongs to the arithmetic, not to the model.*

What each Circle tool does here

Arc Testnet

`eip155:5042002`. The chain settlement lands on. USDC at `0x3600...`, RPC `rpc.testnet.arc.network`.

Circle Gateway

`GatewayClient` holds the buyer's Gateway balance and signs the EIP-3009 authorisation. Deposit, balances and withdrawal all run through it.

Nanopayments x402

`@circle-fin/x402-batching@3.2.0`. Buyer `pay()` runs the full 402 flow against a seller endpoint built with `createGatewayMiddleware`.

x402 facilitator

`gateway-api-testnet.circle.com` verifies and settles. Passed explicitly, because the SDK default is mainnet.

Read from the installed SDK, not the blog post: `GatewayClient` takes a raw EVM key via `viem`. No API key, no entity secret, no Developer-Controlled Wallets in the payment path. That finding made the credential blocker smaller, and it is written down.

SCENARIO C, THE ONE THAT MATTERS

No overlap: both walk away, nothing is paid



WALK-AWAY no payment was made

A deal was never possible: the two owners' limits do not overlap.

BUYER

NO_ZOPA_PRICE

bound by: buyer maximum unit price and total spend cap

final gap 0.000827/call

after 9 rounds

buyer maximum unit price and total spend cap bound after 9 rounds, final gap 827 micro-USDC

SELLER

NO_ZOPA_PRICE

bound by: seller margin floor over cost basis

final gap 0.000827/call

after 9 rounds

seller margin floor over cost basis bound after 9 rounds, final gap 827 micro-USDC

The dashed lines are each side's private limit: neither agent sees the other's, the audience sees both. The seller's floor sits **above** the buyer's ceiling, so no price satisfies both owners. The agents establish that in nine rounds, then each files a post-mortem naming the limit that bound it.

Built, and what is deliberately not claimed

Built

- ✓ Protocol, ledger and a turn loop whose round cap sits outside both agents
- ✓ Guardrail engine: two independent checks over one pure band function
- ✓ Utility, concession schedule, dual ZOPA detection
- ✓ Bounded LLM, wired in, every consultation logged
- ✓ Settlement on accept, post-mortems on walk-away
- ✓ One real payment on Arc Testnet: 9.23 USDC, authorised in 857ms, settled on chain 12m43s later
- ✓ One-screen dashboard, live and replayable

Not claimed

- The real path has run **once**. Every other settlement figure, including all three recorded runs, comes from the **local stub** and is badged **SIMULATED**.
- Authorisation and on-chain settlement are **separate latencies**. Circle batches and there is no flush, so "authorised in 857ms" is not "confirmed in 857ms".
- Testnet only.
- Prompts go to a third-party API.
- No identity or reputation layer. Scoped, then cut on day one of six.